NBA Lower Body Injury Prediction Report

Anthony Seay

April 2026

1 Introduction

In recent years, the NBA (National Basketball Association) has seen a significant increase in lower-body non-contact injuries, such as hamstring, ACL, and Achilles injuries, which are often season-ending for the affected athletes. This project aims to predict these injuries using relevant data such as

1. Player injury history
2. Player tracking data
3. Season schedule

to train multiple models and compare them to see which model performs best at predicting these injuries before the games begin.

1.1 Dataset Specifications

The player statistic data used in the project is pulled directly from the NBA official website via the NBA API, and the injury data is collected from the Kaggle page NBA Injury Stats 1951-2023. The model-ready dataset is constructed from multiple tables collected from the website, sorted by `PLAYER_ID` and `GAME_ID`. This final dataset contains 25,721 observations, each representing a player's performance in a single NBA game, with the target variable `INJURY` being a binary indicator (1 = injury, 0 = no injury). This data is highly imbalanced, with only $\approx 1.19\%$ injury cases, making this a rare-event classification problem and requiring careful evaluation beyond accuracy.

1.2 Feature Overview

The features of the dataset are specifically engineered to capture player workload, fatigue, and recovery, and can be grouped into the following general categories:

- Performance and Tracking Data: Core workload and normalized metrics from in-game tracking.

  - `minutes`, `distance`, `speed`
  - `distance_per_min`, `speed_per_min`, `minutes_prop`

- Rolling Workload Metrics: Short- and long-term averages capturing workload trends over recent games or time windows.
  - `rolling_distance_3`, `rolling_speed_3`, `rolling_minutes_3`
  - `rolling_7d_distance_avg`, `rolling_14d_load_sum`, $\cdots$

- Cumulative and Variability Metrics: Long-term workload and indicators of sudden workload changes.
  - `cumulative_distance`, `cumulative_minutes`
  - `acwr_distance`, `distance_pct_change`, `distance_z`

- Temporal Features: Lagged values and recent game summaries capturing short-term dynamics.
  - `distance_lag1`, `speed_lag1`, `minutes_lag1`
  - `prev_game_minutes`, `prev_3_games_load_sum`

- Recovery and Context Features: Scheduling and game context affecting fatigue and performance.
  - `days_since_prev`, `rest_days`, `back_to_back`
  - `home_game`

- Derived Metrics: Engineered indicators combining workload dimensions.
  - `load_intensity`

These feature groups collectively capture both acute and chronic workload, recent trends, and recovery effects, which are key factors in modeling injury risk making this dataset well suited for logistic regression, gradient boosting, and sequential modeling (RNN's).

2 Methodology and Experimental Evaluation

2.1 Modeling Approach

To predict lower-body non-contact injuries in NBA players, with the help of AI coding assistance I implemented and compared four machine learning models of increasing complexity: Logistic Regression, Gradient Boosting (XGBoost and LightGBM), and a Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM) architecture.

Logistic Regression was used as a baseline model due to its interpretability and widespread use in binary classification tasks. It models the log-odds of

injury occurrence as a linear combination of engineered workload and fatigue-related features. Despite its simplicity, it provides insight into feature importance and serves as a benchmark for more complex models.

Gradient Boosting models (XGBoost and LightGBM) were chosen for their ability to capture nonlinear relationships and feature interactions. These ensemble methods iteratively build decision trees to minimize classification error, making them well-suited for tabular sports data with complex dependencies.

Recurrent Neural Network (RNN) with LSTM units was implemented to explicitly model temporal dependencies in player workload. Unlike the other models, which treat each observation independently, the RNN processes sequences of past games, allowing it to learn patterns related to cumulative fatigue and short-term workload spikes.

2.2 Handling Class Imbalance

Injury prediction is inherently an imbalanced classification problem, with injuries representing a small fraction of observations. To address this, we applied resampling techniques to balance the training data and used evaluation metrics that are robust to class imbalance. In particular, Precision-Recall AUC (PR-AUC) was emphasized over ROC-AUC, as PR-AUC better reflects model performance on rare positive events.

2.3 Evaluation Metrics

Models were evaluated using Accuracy, Precision, Recall, F1 Score, and PR-AUC. While accuracy provides a general measure of correctness, it is not reliable in imbalanced settings. Recall is critical in this application, as missing an injury (false negative) is costly. However, high recall alone is insufficient if it leads to excessive false positives. Therefore, F1 Score and PR-AUC were used to assess the trade-off between precision and recall.

2.4 Results

Model	Accuracy	Precision	Recall	F1 Score	PR-AUC
Logistic Regression	0.44	0.02	**0.98**	0.04	0.105
XGBoost	0.72	0.03	**0.77**	0.06	0.043
LightGBM	0.74	0.03	**0.74**	0.06	0.045
RNN (LSTM)	**0.82**	**0.18**	0.62	**0.28**	**0.198**

Table 1: Model performance comparison on test data

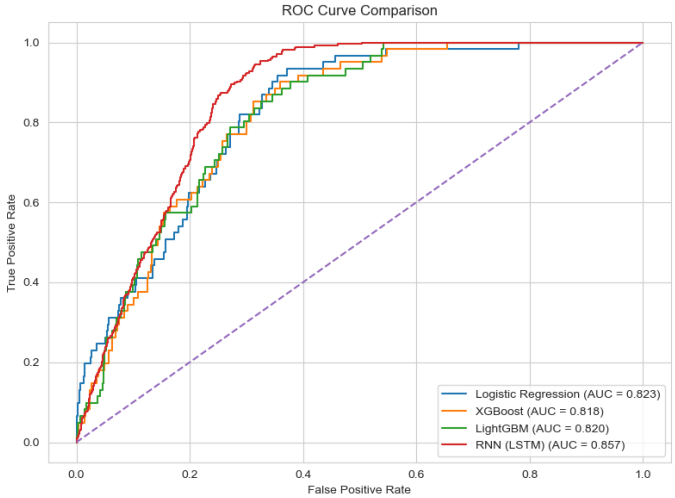


Figure 1: ROC curve comparison of all models. While all models achieve relatively high ROC-AUC scores, the RNN performs the best in this metric

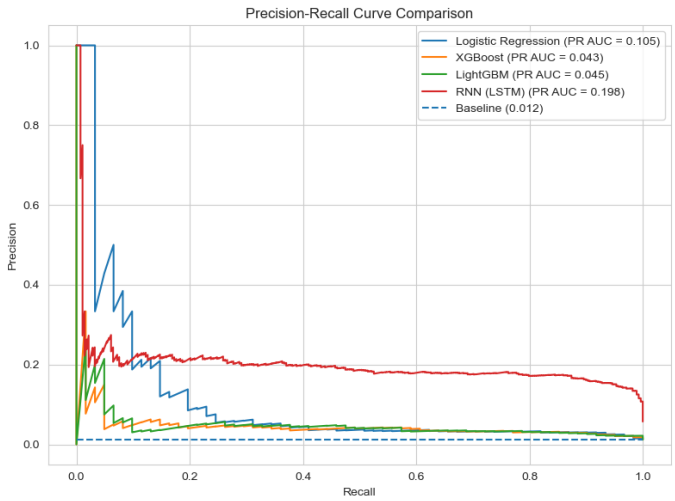


Figure 2: Precision-Recall is a better metric too evaluate these models due the class imbalance of the data set. The RNN model achieves the highest PR-AUC, demonstrating superior performance in identifying rare injury events under class imbalance. The dashed line represents the baseline positive class rate.

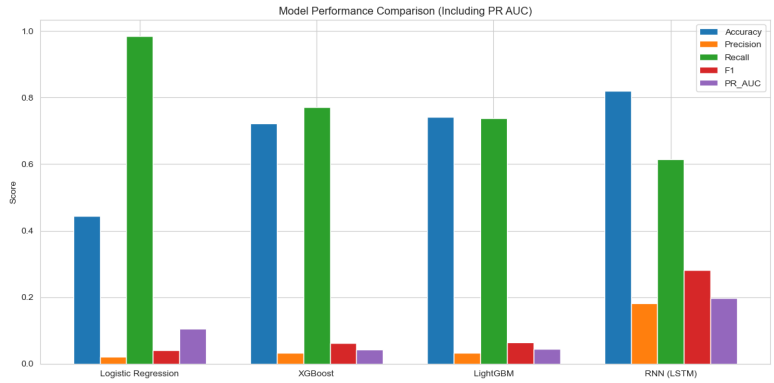


Figure 3: Comparison of model performance metrics. The RNN achieves the best balance between precision and recall, while Logistic Regression achieves high recall at the cost of extremely low precision.

2.5 Discussion

The results highlight several important insights about injury prediction.

The Logistic Regression model achieves extremely high recall (0.98), meaning it successfully identifies nearly all injury cases. However, its precision is extremely low (0.02), indicating that the vast majority of predicted injuries are false positives. This behavior reflects an over-prediction tendency, making the model impractical despite its strong recall.

The Gradient Boosting models (XGBoost and LightGBM) improve overall accuracy and maintain relatively high recall (0.77 and 0.74, respectively). However, their precision remains very low (0.03), and PR-AUC scores are significantly lower than Logistic Regression. This suggests that while these models better separate classes globally, they still struggle to identify true injury cases effectively under class imbalance.

The RNN (LSTM) model demonstrates the best overall performance across nearly all metrics. It achieves the highest accuracy (0.82), significantly higher precision (0.18), and the strongest PR-AUC (0.198). Although its recall (0.62) is lower than other models, it achieves a much better balance between detecting injuries and limiting false positives. This indicates that incorporating temporal information through sequence modeling provides a meaningful advantage in capturing injury risk patterns.

2.6 Limitations and Error Analysis

Despite improvements with the RNN, several limitations remain.

First, the dataset is highly imbalanced, which inherently limits precision across all models. Even the best-performing model (RNN) produces a nontrivial number of false positives, which may reduce its practical utility in real-world decision-making.

Second, the feature set, while extensive, may not fully capture all relevant factors influencing injury risk, such as biomechanical data, player health history beyond recorded injuries, or contextual variables like travel and recovery conditions.

Third, the RNN model, while more powerful, is less interpretable than Logistic Regression and tree-based models. This creates a trade-off between predictive performance and explainability, which is important in sports analytics applications where decisions must be justified.

Finally, threshold selection plays a critical role in model performance. While thresholds were optimized using F1 score, different operational settings (e.g., prioritizing recall over precision) may require alternative threshold strategies.

2.7 Summary

Overall, this experiment creates a great starting point for injury prevention in basketball using machine learning techniques. With further improvements in dataset availability and feature engineering this sort of model can be very useful in managing player workload.